

Rahul Reddy Kesari

Denton, TX

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CCNA-certified Network Engineer with 2 years of hands-on experience supporting, configuring, monitoring, and troubleshooting enterprise network infrastructure. Skilled in Cisco routing and switching, TCP/IP, IPv4/IPv6, VLANs, trunking, inter-VLAN routing, OSPF, EIGRP, BGP fundamentals, STP/RSTP, EtherChannel, DHCP, DNS, NAT/PAT, ACLs, VPN fundamentals, and network security. Experienced in diagnosing connectivity, routing, VLAN, interface, and performance issues using Cisco IOS, Wireshark, GNS3, Linux, ping, traceroute, routing tables, logs, and packet analysis. Familiar with Python, Netmiko, Ansible, REST APIs, Git/GitHub, Aruba, Juniper, Palo Alto NGFW, AWS networking, network monitoring, incident management, and automation workflows. Seeking Network Engineer, NOC Engineer, Network Support Engineer, or Infrastructure Engineer opportunities.

CERTIFICATIONS:

- **Cisco Certified Network Associate (CCNA) - Cisco | Aug 2026 – Aug 2029**
Credential ID: CSC015220767
- **Crash Course on Python - Google**
- **Tools for Data Science - Coursera**

TECHNICAL SKILLS:

Routing & Switching: Cisco IOS, IPv4/IPv6, Subnetting, Static Routing, OSPF, EIGRP, BGP Fundamentals, VLANs, 802.1Q, Inter-VLAN Routing, STP/RSTP, EtherChannel/LACP, HSRP, VRRP

Network Services & Security: DHCP, DNS, NTP, SNMP, Syslog, SSH, NAT/PAT, ACLs, Port Security, IPsec VPN, Firewall Fundamentals

Monitoring & Troubleshooting: Wireshark, Ping, Traceroute, MTR, Packet Analysis, Routing/MAC/ARP Table Analysis, Log Analysis, Fault Isolation, Root Cause Analysis

Network Automation: Python, Netmiko, Ansible Fundamentals, REST API Fundamentals, JSON/YAML, Git/GitHub, Linux

Cloud Networking: AWS VPC/Subnets/Route Tables/Security Groups Fundamentals, Junos Fundamentals, Aruba Fundamentals, Palo Alto NGFW Fundamentals

Linux: Linux CLI, SSH, Network Configuration, Routing, DNS Troubleshooting, Permissions, Process & Log Analysis

Tools: Cisco Packet Tracer, GNS3, EVE-NG, Wireshark, MobaXterm, PuTTY, GitHub, AWS Console

EDUCATION:

M.S., Computer and Information Sciences

University of North Texas, Denton, TX, USA

GPA: 3.9/4.0

08/2024 – 05/2026

Relevant Coursework:

- Networks & Communications: Computer Networks, Wireless Networks
- Systems & Software: Software Engineering, Graph Theory
- Multimedia & Signals: Multimedia Computing

B.Tech., Electronics and Communication Engineering

S.R.M Institute of Science and Technology, Kattankulathur, TamilNadu, India

GPA: 9.04/10.0

09/2020 – 05/2024

Relevant Coursework:

- Communication & Networks: Computer Networks, Data Communications, Wireless Communication
- Signals & Imaging: Signal Processing, Digital Imaging
- Digital & VLSI Systems: Digital Circuit Design, Advanced Digital System Design, VLSI Design
- Control & Electronics: Control Systems, Analog Circuits

WORK EXPERIENCE:

Network Engineering

Bharat Sanchar Nigam Limited (BSNL) | India

Aug 2022

- Assisted with telecom network operations involving switching, transmission systems, and network infrastructure.
- Gained exposure to wireline and wireless communication systems, signal transmission, and fault monitoring.
- Supported routine network monitoring, fault identification, maintenance, and operational workflows in a production telecom environment.
- Observed troubleshooting and maintenance procedures used to support network availability and service continuity.

PROJECTS:

Enterprise Campus Network Design & High Availability

Cisco IOS | OSPF | VLANs | STP/RSTP | EtherChannel | HSRP | DHCP | NAT | ACLs

- Designed a multi-segment enterprise network with Cisco routers and switches, implementing VLAN segmentation, 802.1Q trunking, inter-VLAN routing, and structured IPv4 addressing.
- Implemented OSPF dynamic routing, STP/RSTP, EtherChannel, and HSRP to provide routing convergence and Layer 2/3 redundancy.
- Configured DHCP, NAT/PAT, ACLs, SSH, and port security to support network services and controlled access.
- Simulated link, interface, VLAN, and routing failures and performed fault isolation using Cisco IOS show commands, ping,

tracert, and routing/MAC/ARP table analysis.

- Validated end-to-end connectivity, redundant-path operation, and authorized inter-VLAN communication across the lab environment.

Multi-ISP WAN Routing & Failover

GNS3 | Cisco IOS | OSPF | BGP | Static Routing | IP SLA

- Built a multi-router WAN topology representing enterprise edge and multiple ISP networks using GNS3.
- Configured OSPF, BGP fundamentals, static/default routing, and redundant WAN paths for internal and external route exchange.
- Troubleshoot missing routes, incorrect next hops, return-path failures, OSPF adjacencies, and BGP prefix advertisements.
- Verified route selection and end-to-end connectivity using show ip route, show ip ospf neighbor, show ip bgp, ping, and tracert.

Network Automation & Configuration Management

Python | Netmiko | Ansible | REST APIs | JSON/YAML | Git/GitHub

- Developed Python and Netmiko workflows to establish SSH sessions and execute operational commands across simulated network devices.
- Automated collection of interface status, routing information, VLAN data, device inventory, and running configurations.
- Used YAML-based device inventories and Ansible fundamentals to perform repeatable multi-device configuration tasks.
- Implemented automated configuration backup, validation, exception handling, and logging workflows.
- Managed automation scripts and configuration templates with Git/GitHub for version control and change tracking.

Palo Alto NGFW & Site-to-Site VPN Lab

Palo Alto PA-VM | Security Policies | NAT | IPsec | GlobalProtect | Wireshark

- Built a firewall lab using Palo Alto PA-VM with Layer 3 interfaces, security zones, routing, and security policies.
- Configured source NAT and policy-based traffic controls between trusted, untrusted, and internal network segments.
- Implemented an IPsec site-to-site VPN using IKE Phase 1 and IPsec Phase 2 parameters and validated encrypted connectivity.
- Tested URL filtering, security policy matching, logging, and traffic-session analysis. • Troubleshoot policy, NAT, routing, and VPN connectivity issues using firewall logs, session information, ping, tracert, and packet analysis.

Network Monitoring & Incident Troubleshooting

Wireshark | SNMP | Syslog | TCP/IP | DNS | DHCP | Troubleshooting

- Simulated enterprise network incidents involving interface failures, VLAN/trunk issues, routing problems, DHCP failures, and DNS reachability.
- Captured and analyzed ARP, ICMP, DHCP, DNS, TCP, and UDP traffic using Wireshark to isolate protocol and connectivity issues.
- Correlated interface status, routing tables, MAC/ARP tables, logs, and packet captures to identify root causes.
- Analyzed TCP three-way handshakes, retransmissions, DHCP DORA exchanges, DNS queries, and ICMP traffic during troubleshooting.
- Documented incident symptoms, troubleshooting actions, root cause, resolution, and validation steps using an RCA-style workflow.

PUBLICATIONS:

Enhanced Gesture Recognition Using Hand Gesture and Text Integration

Python | OpenCV | Image Processing

2024

- Developed a Python and OpenCV-based gesture recognition system capable of detecting and classifying 16 hand gestures in real time.
- Achieved 92% gesture-recognition accuracy by integrating image-processing techniques with text-based interpretation.
- Evaluated the system across 500+ gesture samples to measure recognition accuracy and improve classification reliability.